

DoRA the EX-plorer

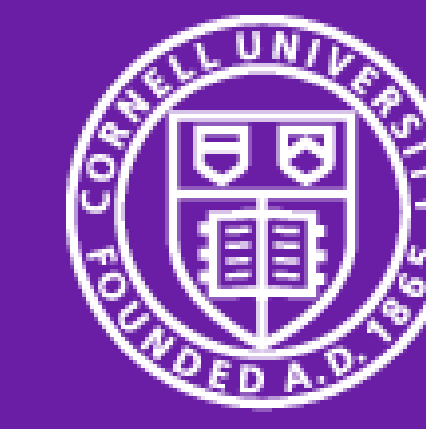
An Adventure in Parameter-Efficient Fine-Tuning with DoRA, the supportive EX-plorer

Keng-Lien Lin

Yiyi Wu

Ellen Foreman

CS 5782



CornellBowers
Computer Science

01 – Introduction & Background

Fine-tuning is expensive!

*PEFT: Parameter Efficient Fine Tuning

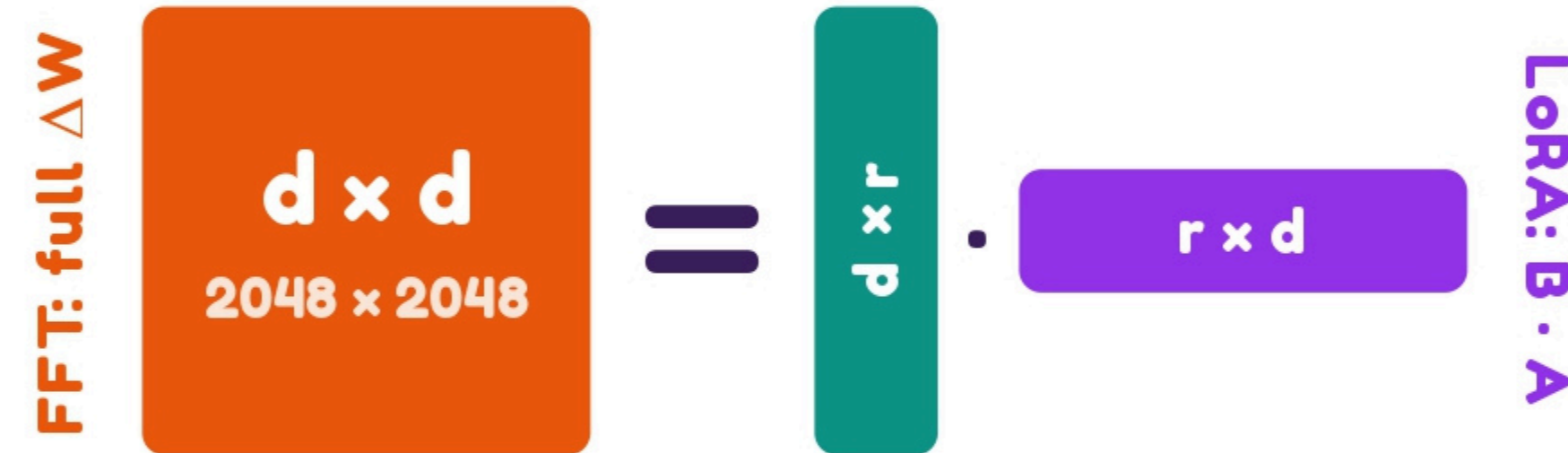
Modern LLMs have billions of parameters. Full fine-tuning requires updating *every* weight — impractical for most researchers. We investigate **DoRA**, a PEFT method that decomposes weight updates into **magnitude & direction** for better efficiency.

Full Fine-Tuning (FFT)

Budget: d^2

$$W' = W_0 + \Delta W; \quad \Delta W = -\eta \nabla_W \mathcal{L}$$

For Llama 3.2 1B, $d = 2048 \rightarrow$ one square layer $\approx$ **4.2M params**.



Low-Rank Adaptation (LoRA)

Budget: $2dr$

$$B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times d}, \quad \Delta W = \frac{\alpha}{r} BA$$

Freezes the pretrained weight matrix W_0 and learns low-rank updates

$$W' = \|W_0 + sBA\|_{\text{row}} \cdot \frac{W_0 + sBA}{\|W_0 + sBA\|_{\text{row}}}$$

Magnitude & direction are **entangled** — no independent control.

Weight-Decomposed LoRA (DoRA)

Budget: $2dr + d$

$$W' = m \cdot \frac{W_0 + sBA}{\|W_0 + sBA\|_{\text{row}}}, \quad m \in \mathbb{R}^{d \times 1}$$

- Decomposes weights into magnitude (m) and direction (V).
- Mimics Full Fine-Tuning (FFT) optimization trends better than LoRA.

Sources

- [1] Liu et al. (2024). *DoRA: Weight-Decomposed Low-Rank Adaptation*. arXiv:2402.09353
- [2] Hu et al. (2022). *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR 2022.
- [3] Meta AI (2024). *Llama 3.2*. HuggingFace.

02 – Methodology

Configuration

Base model	Llama 3.2 1B (Transformer LLM)
Benchmark	Commonsense Reasoning QA — BoolQ, Hella., Wino., ARC-e/c, OBQA, PIQA, SIQA
Dataset	Commonsense Reasoning 170K
Target Modules	Q, K, V, Up, Down
Variants	3 seed $\times$ ranks [8, 4, 2] for LoRA & DoRA

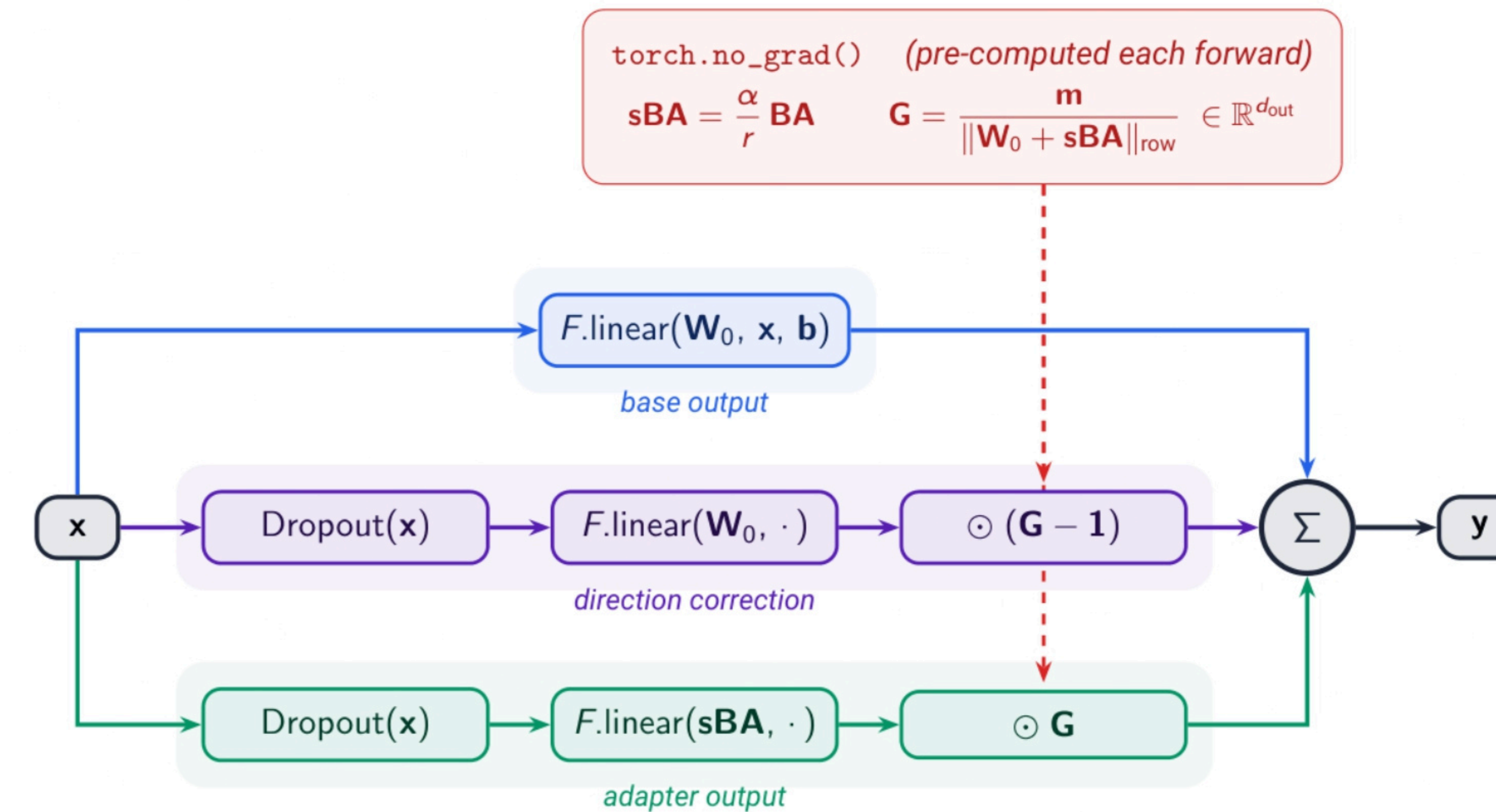
★ DoRA structure validated against the official DoRA library before experiments

Efficient Computation Trick - Forward Pass

Naively computing $Y = G(W_0 + sBA)x$ is expensive

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let  $G = m / \|W_0 + sBA\|_{\text{row}}$   
→  $W_0x - W_0x + G(W_0 + sBA)x$   
→  $W_0x + (G-1)W_0x + G(sBA)x$ 
```

Avoids computing W' directly, enabling **mixed-precision efficiency**



WHAT WE DID DIFFERENTLY

- Minimal performance gap at rank = 16, likely capacity saturation
- Ranks [8, 4, 2] — appropriate for smaller 1B model
- 20K training slice due to compute constraints
- 3 seeded runs per variant for statistical stability

FUTURE DIRECTIONS

- Investigate magnitude-aware PEFT methods (MAP)
- Explore domain beyond text, e.g. Vision, Audio

03 – Results & Conclusion

Magnitude & Direction Relationship

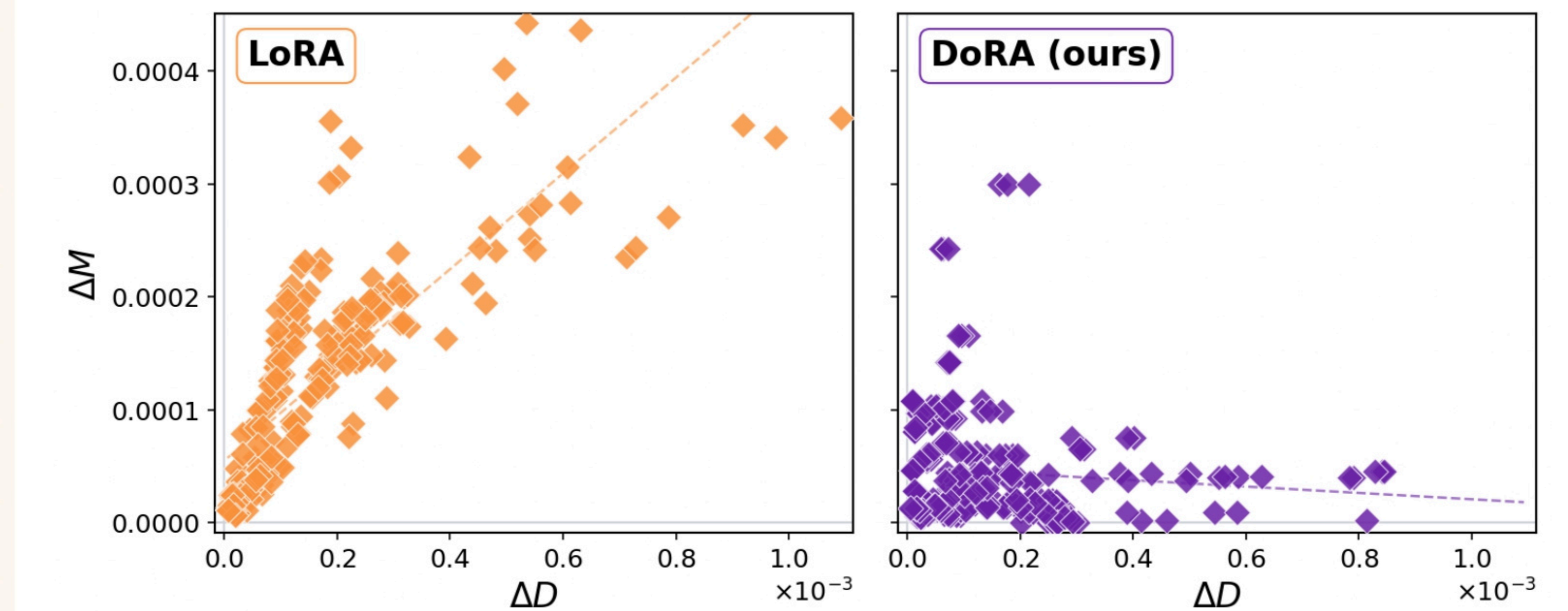


Figure 1. Magnitude and direction updates for target modules. Demonstrates distinct correlation patterns, highlighting DoRA's ability to decouple updates.

LoRA vs DoRA on Commonsense QA

Method	Rank	Params%	BoolQ	Hella.	Wino.	ARC-e	ARC-c	OBQA	PIQA	SIQA	Avg.
Base	—	0.000%	37.8	44.6	51.6	56.6	35.8	31.6	78.2	16.2	44.1
LoRA	8	0.309%	37.4	46.8	50.2	63.7	39.5	34.2	74.7	16.5	45.4
DoRA	8	0.326%	37.8	46.8	52.4	64.4	37.4	33.3	75.1	16.1	45.4
DoRA(lib)	8	0.326%	38.7	46.3	50.6	63.3	39.8	35.0	75.3	16.3	45.7
LoRA	4	0.155%	37.6	46.1	50.0	63.3	37.2	33.5	75.2	16.8	45.0
DoRA	4	0.172%	38.5	46.4	50.7	63.9	35.9	33.7	75.1	16.8	45.1
LoRA	2	0.077%	38.4	45.9	50.1	62.2	34.8	32.4	74.8	16.7	44.4
DoRA	2	0.095%	38.6	46.4	50.2	63.8	35.3	33.5	75.0	16.8	45.0

Table 1. Accuracy comparison of Llama 3.2 1B with LoRA and DoRA on CommonsenseQA

Average Accuracy vs Rank

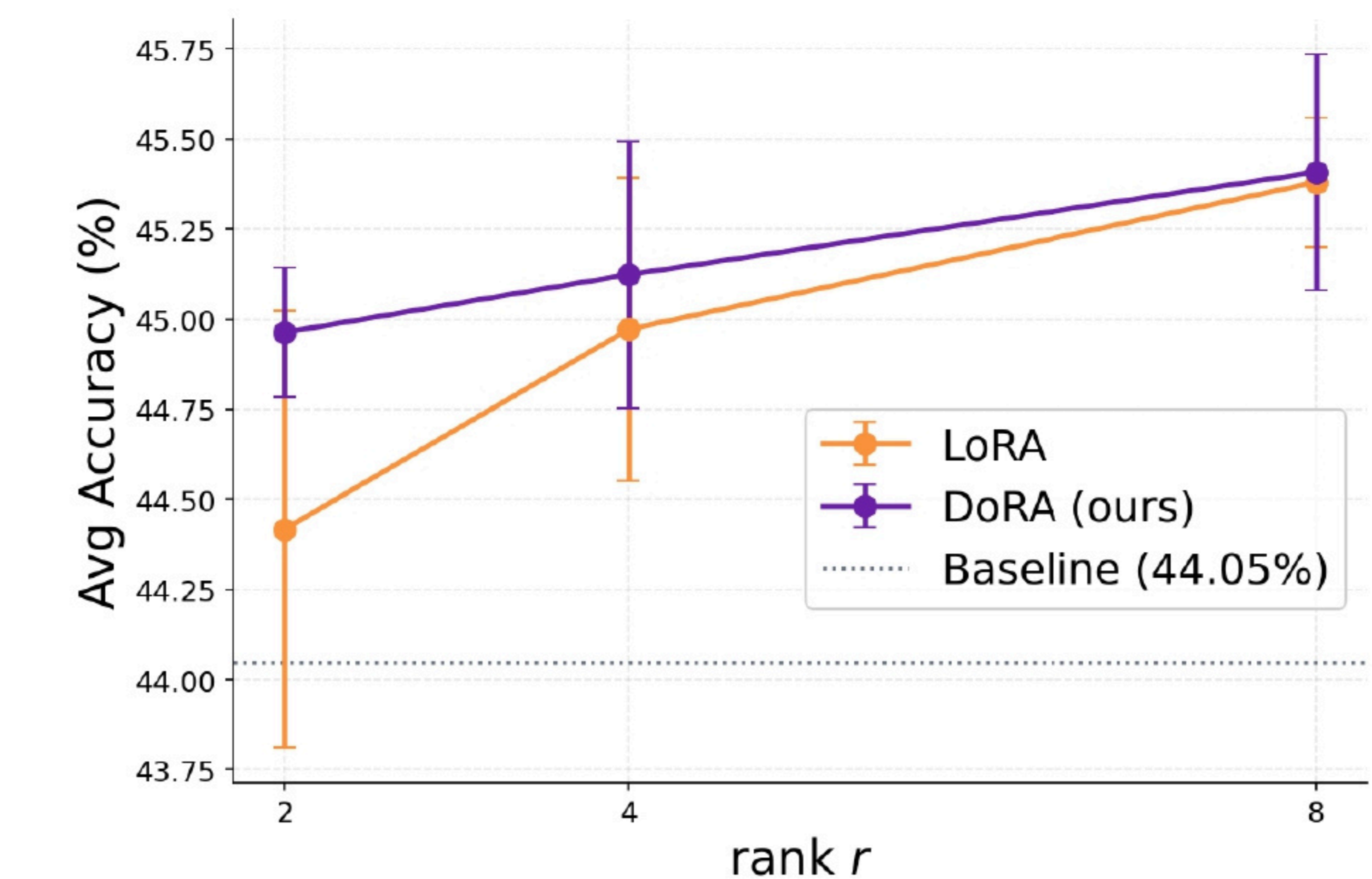


Figure 2. Average accuracy of LoRA, DoRA, and baseline across rank = 2, 4, and 8.

Key Takeaways

- DoRA **consistently outperforms** LoRA across all ranks
- DoRA adds only d extra params over LoRA — negligible overhead
- Gains persist at very low rank ($r = 2$) where LoRA degrades more
- Drop-in replacement for LoRA with no architecture change required